Nicolás Francinelli

Robotics Engineer & Cluster Leader | ROS 2 · C++ · Python

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PROFESSIONAL EXPERIENCE

ALTEN – cluster of 5 consultants on Airbus projects — Toulouse, France
Cluster Leader – Maintenance (concurrent role) — Oct 2026 – Present

- Lead 5 ALTEN engineers: task follow-up, reporting to management and early escalation of risks.
- Identify needs from engineers' feedback and propose improvements and new initiatives to Airbus.

ALTEN – client: Airbus (OMR) — Toulouse, France
Robotics Engineer (Consultant) — Mar 2026 – Present

Primer Structural Defects project (~20 people): automated fuselage defect inspection replacing manual checks.

- Sole developer of the Siemens Industrial PC software, including TIA Portal and PROFINET setup with the PLC.
- Implement a 30–40-node Behavior Tree process controller handling 7–8 HMI message types over a TCP/IP server and 10–15 PLC signals through shared memory.

LR Technologies – client: Magellium, MMX space mission (CNES) — Toulouse, France
Robotics Engineer (Consultant) — Sep 2025 – Dec 2025

- Built an automated validation framework in C and Python for test campaigns on 3 platforms, cutting a 2-week manual process to 3–4 hours plus results analysis.
- Automated code-coverage analysis, memory analysis, log parsing and results visualization.
- Wrote 70+ unit tests (CUnit) and improved software quality and technical documentation.

Illumo Robotics – industrial AI-based pick-and-place startup — Toulouse, France
Robotics Engineer — Dec 2023 – May 2025

- Owned path planning, simulation, sensor integration (team of 4); helped cut pick cycle time from 3–4 s to 2 s.
- Migrated ~10 packages from ROS to ROS 2 and designed ROS 2 nodes for robot control, trajectory planning and behavior management (state machines, behavior trees).
- Developed Gazebo plugins for a Yaskawa industrial robot arm and its actuators, enabling simulation-first development and bug-fix verification before deployment to the test cell.
- Integrated industrial hardware: FRAMOS Ethernet camera and IFM IO-Link master.
- Software deployed on 2 robot cells for a customer in Spain; delivered 5+ robot safety training sessions to six-person teams on site.

Capgemini Engineering — Toulouse, France
R&D Robotics Engineer Intern — Feb 2023 – Aug 2023

- Simulated a mobile robot for EV charging with autonomous navigation and dynamic obstacle avoidance.
- Estimated object position from camera data and implemented admittance control on a UR10e robot arm.

EDUCATION

École Centrale de Nantes — Nantes, France
MSc in Advanced Robotics (CORO-IMARO) — Sep 2021 – Aug 2023

- GPA 3.49/4. Student representative of the Control and Robotics Master's program for 2 years.

Universidad Nacional de La Plata (UNLP) — La Plata, Argentina
Electronics Engineering Degree — Mar 2016 – Feb 2021

- Average 8.77/10; Best Graduate 2020; Teaching Assistant in Circuit Theory I for 2.5 years.
- Final project (LEICI lab): CNN-based autonomous navigation, odometry/IMU Kalman filter, PID heading control.

TECHNICAL SKILLS

- **Programming & Tools:** C, C++, Python, Linux, Git, Docker, CAD
- **Robotics:** ROS, ROS 2, Gazebo, Behavior Trees, State Machines, Trajectory Planning, Autonomous Navigation, Admittance Control, PID Control
- **Industrial Automation:** Siemens TIA Portal, Siemens Industrial PC, PLC, PROFINET, HMI, TCP/IP, IO-Link
- **Perception & AI:** OpenCV, Computer Vision, Machine Learning, CNN, Kalman Filter, Sensor Fusion
- **Testing & Quality:** Unit Testing (CUnit), Code Coverage, Memory Analysis, Test Automation

LANGUAGES & CERTIFICATIONS

- **Languages:** Spanish (native), English (C2), French (C2), Italian (A2)
- **Certifications:** CodinGame Certification – C++ (2022)
- **Interests:** Cooking, rugby, Formula 1